

```
import numpy as np
import pandas as pd
```

```
df= pd.read_csv('customer_shopping_behavior.csv')
```

```
df.head()
```

	customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rati
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                   3900 non-null   int64
2   Gender                               3900 non-null   object
3   Item Purchased                       3900 non-null   object
4   Category                             3900 non-null   object
5   Purchase Amount (USD)                3900 non-null   int64
6   Location                             3900 non-null   object
7   Size                                  3900 non-null   object
8   Color                                3900 non-null   object
9   Season                                3900 non-null   object
10  Review Rating                         3863 non-null   float64
11  Subscription Status                   3900 non-null   object
12  Shipping Type                         3900 non-null   object
13  Discount Applied                     3900 non-null   object
14  Promo Code Used                       3900 non-null   object
15  Previous Purchases                    3900 non-null   int64
16  Payment Method                       3900 non-null   object
17  Frequency of Purchases                3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
df.describe()
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases	
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000	
mean	1950.500000	44.068462	59.764359	3.750065	25.351538	
std	1125.977353	15.207589	23.685392	0.716983	14.447125	
min	1.000000	18.000000	20.000000	2.500000	1.000000	
25%	975.750000	31.000000	39.000000	3.100000	13.000000	
50%	1950.500000	44.000000	60.000000	3.800000	25.000000	
75%	2925.250000	57.000000	81.000000	4.400000	38.000000	
max	3900.000000	70.000000	100.000000	5.000000	50.000000	

```
df.describe(include='all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Sub
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	

df.isnull().sum()

	0
Customer ID	0
Age	0
Gender	0
Item Purchased	0
Category	0
Purchase Amount (USD)	0
Location	0
Size	0
Color	0
Season	0
Review Rating	37
Subscription Status	0
Shipping Type	0
Discount Applied	0
Promo Code Used	0
Previous Purchases	0
Payment Method	0
Frequency of Purchases	0

dtype: int64

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(
    lambda x: x.fillna(x.median())
)
```

df.isnull().sum()

	0
Customer ID	0
Age	0
Gender	0
Item Purchased	0
Category	0
Purchase Amount (USD)	0
Location	0
Size	0
Color	0
Season	0
Review Rating	0
Subscription Status	0
Shipping Type	0
Discount Applied	0
Promo Code Used	0
Previous Purchases	0
Payment Method	0
Frequency of Purchases	0

dtype: int64

```
df.columns = df.columns.str.lower()
df.columns= df.columns.str.replace(' ', '_')
df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'}, inplace=True)
```

df.columns

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

```
labels= ['young adult', 'adult', 'middle_aged', 'senior']
df['age_group']= pd.qcut(df['Age'],q=4, labels=labels)
```

df[['age_group', 'Age']].head()

	age_group	Age
0	middle_aged	55
1	young adult	19
2	middle_aged	50
3	young adult	21
4	middle_aged	45

```
# Create mapping dictionary
frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}

# Create new column
df['purchase_frequency_days'] = df['Frequency of Purchases'].map(frequency_mapping)
```

```
df[['purchase_frequency_days', 'Frequency of Purchases']].head()
```

	purchase_frequency_days	Frequency of Purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually

```
df[['discount_applied', 'promo_code_used']].head()
```

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes

```
(df['discount_applied'] == df['promo_code_used']).all()
```

```
np.True_
```

```
if 'promo_code_used' in df.columns:  
    df = df.drop('promo_code_used', axis=1)
```

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',  
      'purchase_amount', 'location', 'size', 'color', 'season',  
      'review_rating', 'subscription_status', 'shipping_type',  
      'discount_applied', 'previous_purchases', 'payment_method',  
      'frequency_of_purchases'],  
      dtype='object')
```

```
df.to_csv('cleaned_customer_shopping_behavior.csv', index=False)
```

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